ECE 447

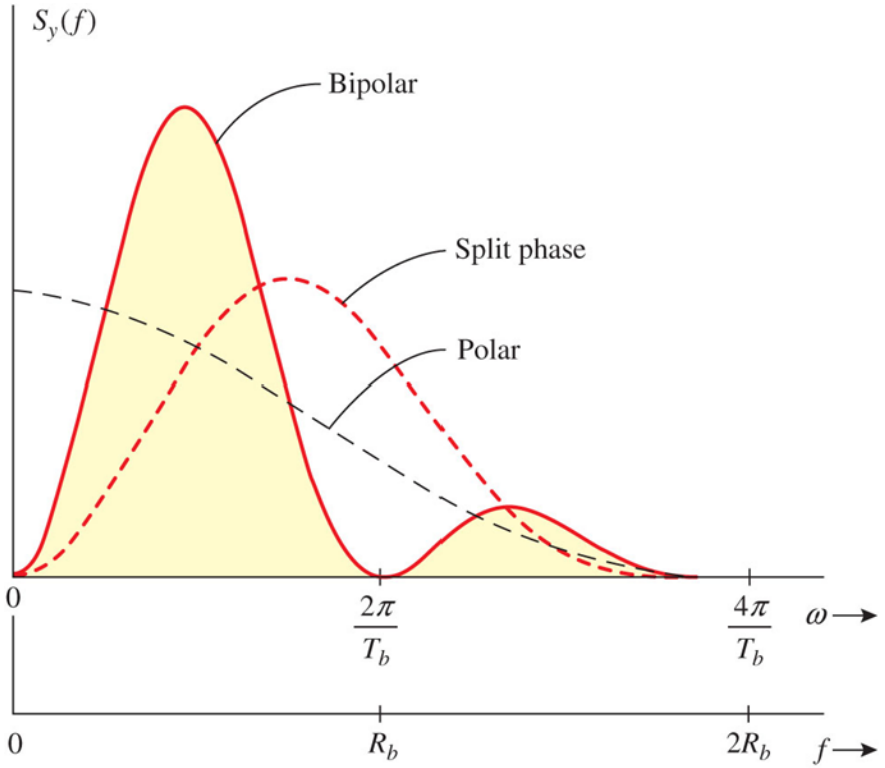
Lesson 27
Digital Carrier Modulation and Demodulation (Bandpass Signals)



Schedule and Admin

- Schedule
- Admin
  - **Lab 5**. Due Lesson 27 (today) to GradeScope
  - **HW6**. Due Lesson 28 (Tue, 27 Oct) to GradeScope.
  - **HW7**. Due Lesson 32 (Fri, 6 Nov) to GradeScope. Today covers Problem 4.
  - **Lab 6**. Begins Lesson 28 (Tue, 27 Oct), runs Lessons 28-29; PDF due Lesson 31 (Wed, 4 Nov).

REVIEW



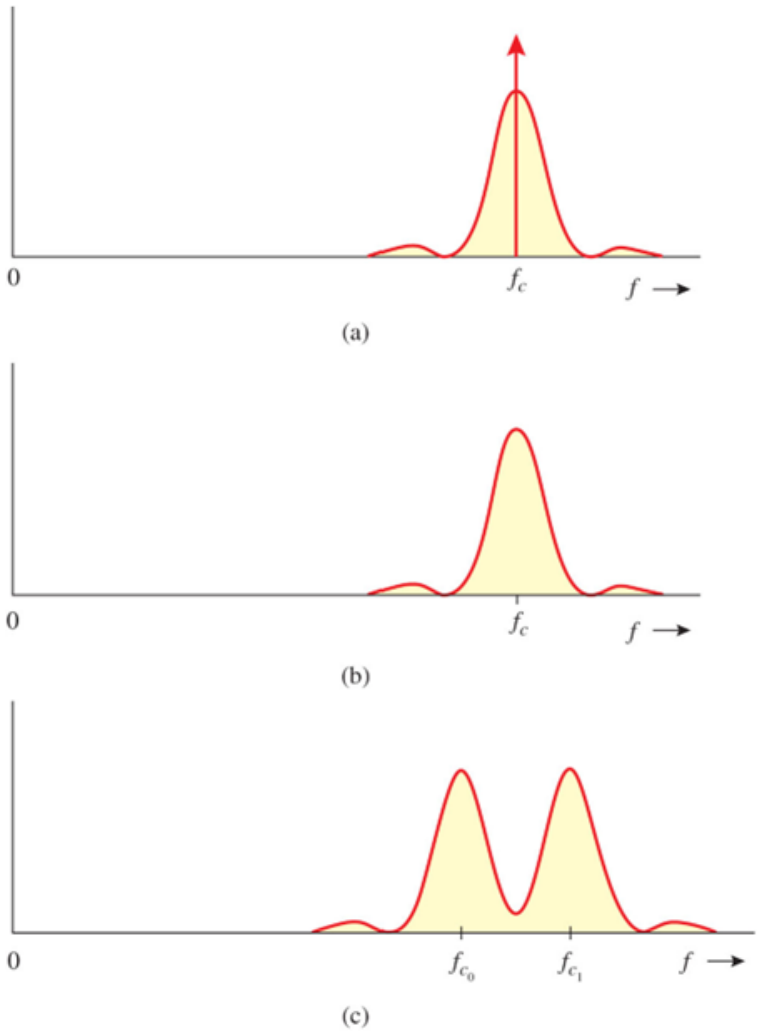
- The baseband polar signaling BW in this figure is for RZ. If changed to NRZ (full-width pulse), the bandwidth reduces to R_b. (Important for HW 7, Problem 4 – Lathi 6.8-1)

Binary Carrier Modulations

- ASK - self-explanatory
- Angle modulations: PSK and FSK
- PSK
  - If $m(t)$ uses polar signaling $\{\pm 1\}$, phase shifts for BPSK signal, $\varphi_{PSK}(t)$, is π radians apart
  - $\varphi_{PSK}(t) = m(t)\cos(\omega_c t)$
  - Constellation plot?
- FSK:
  - Sum of two alternating ASK signals with different carrier frequencies, ω_{c0} and ω_{c1}
  - $a_k = \{0, 1\}$
  - $\varphi_{FSK}(t) = \Sigma a_k p(t - kT_b)\cos(\omega_{c1} t) + \Sigma(1 - a_k)p(t - kT_b)\cos(\omega_{c0} t)$
- FT modulation property derives PSDs of digital bandpass signals:

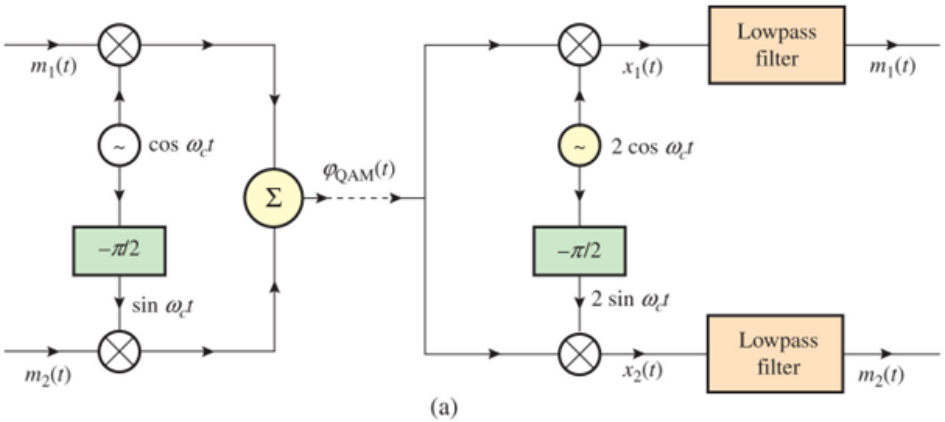
$$S_\varphi(f) = \frac{1}{4}S_M(f + f_c) + \frac{1}{4}S_M(f - f_c)$$

Binary Carrier Modulations

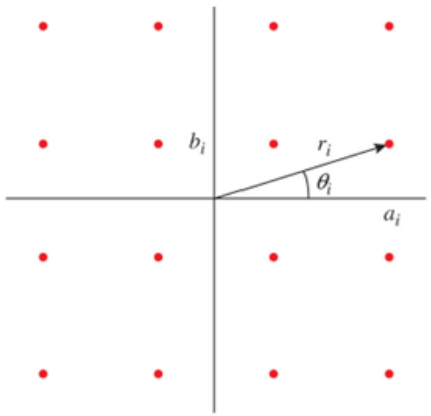


M-ary Carrier Modulations

- Focus on QAM - most widely used in modern systems



(a)



(b)